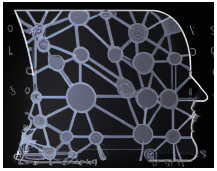
Applied Machine Learning

Feature Engineering 2 - Feature Selection

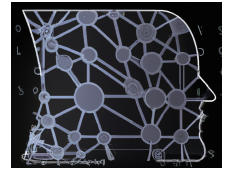


Learning goals

- Reasons for Feature Selection
- Types of Feature Selection

Recap

WHAT IS FEATURE ENGINEERING?



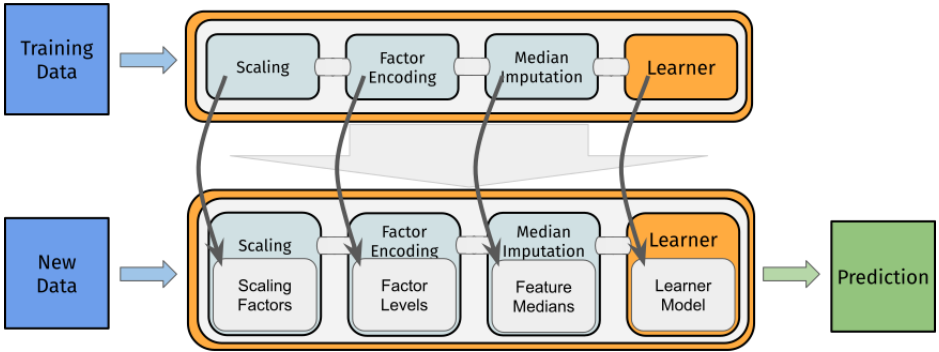
Feature engineering is on the intersection of **data cleaning**, **feature creation** and **feature selection**.

The goal is to solve common difficulties in data science projects, like

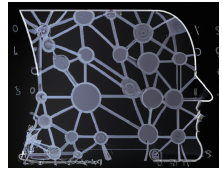
- skewed/weird feature distributions,
- (high cardinality) categorical features,
- functional (temporal) features,
- missing observations,
- high dimensional data,
- ...

and improve model performance (or make the data readable by the model in the first place).

TRAIN-TEST LEAKAGE



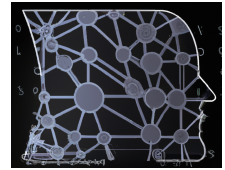
TYPES OF FEATURE ENGINEERING (SO FAR)



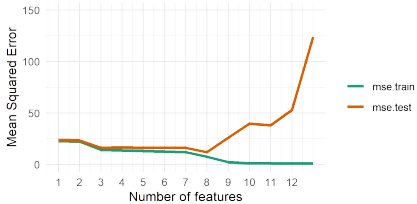
- numerical encoding of categorical features
- feature transformations
- imputation

Feature Selection

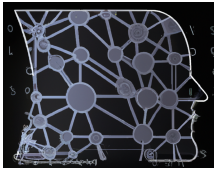
MOTIVATION



- Naive view:
  - More features $\rightarrow$ more information $\rightarrow$ discriminant power $\uparrow$
  - Model is not harmed by irrelevant features since their parameters can simply be estimated as 0.
- In practice, irrelevant and redundant features can "confuse" learners (see **curse of dimensionality**) and worsen performance.
- Example: In linear regression, R^2 is monotonically increasing in p, but adding irrelevant features leads to overfitting (capturing noise).

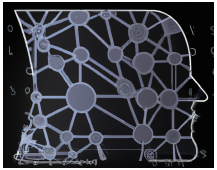


MOTIVATION



- In high-dimensional data sets, we often have prior information that many features are either irrelevant or of low quality
- Having redundant features can cost something during prediction (money or time)
- Many models require $n > p$ data. Thus, we either need to
  - adapt models to high-dimensional data (e.g., regularization)
  - design entirely new procedures for $p > n$ data
  - use filter preprocessing methods from this lecture

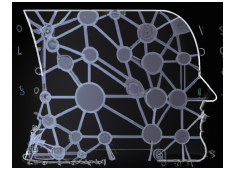
SIZE OF DATASETS



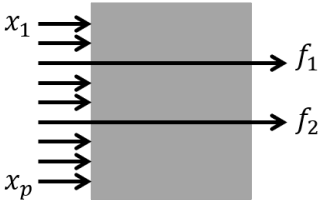
Many new forms of technical measurements and connected data leads to availability of extremely high-dimensional data sets.

- **Classical setting**: Up to around 10^2 features, feature selection might be relevant, but benefits often negligible.
- **Datasets of medium to high dimensionality**: At around 10^2 to 10^3 features, classical approaches can still work well, while principled feature selection helps in many cases.
- **High-dimensional data**: 10^3 to 10^9 or more features. Examples: micro-array / gene expression data and text categorization (bag-of-words features). If we also have few observations, scenario is called $p \gg n$.

FEATURE SELECTION VS. EXTRACTION

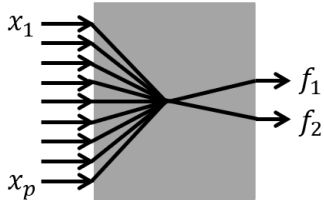


Feature selection



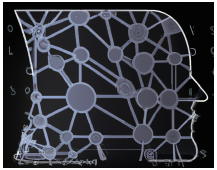
- Creates a subset of original features **x** by selecting $\tilde{p} < p$ features ***f***.
- Retains information on selected individual features.

Feature extraction



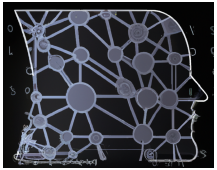
- Maps p features in **x** to $\tilde{p}$ extracted features ***f***.
- Info on individual features can be lost through (non-)linear combination.

REASONS FOR FEATURE SELECTION



- improved predictive performance, since we reduce overfitting on irrelevant features,
- robust models that do not rely on noisy features,
- simpler models that are easier to interpret,
- faster model fitting, e.g. for model updates,
- faster prediction, and
- no need to collect potentially expensive features.

GOALS OF FEATURE SELECTION

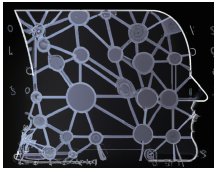


Single Feature Selection Problem: identifying a minimal-size subset of the variables that is optimally predictive for an outcome variable T of interest.

Multiple Feature Selection Problem: Let $\mathcal{S}$ be the solution to the single feature selection problem (called the reference solution). The solution $\mathcal{M}$ to the multiple feature selection problem consists of all minimal-size sets $\mathcal{S}_i \in \mathcal{F}$ that are statistically equivalent to $\mathcal{S}$.

- Important for knowledge discovery (know all possible solutions)
- We will not delve into this any further

TYPES OF FEATURE SELECTION METHODS

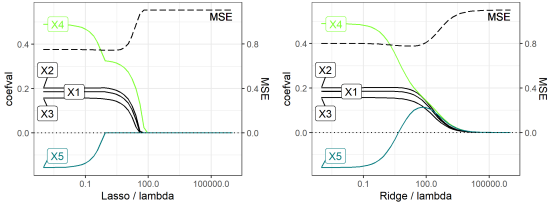


In rest of the chapter, we introduce different types of methods for FS:

- Filters: evaluate relevance of features using statistical properties such as correlation with target variable
- Wrappers: use a model to evaluate subsets of features
- Embedded methods: integrate FS directly into specific model - we look at them in their dedicated chapters (e.g., CART, L_0, L_1)

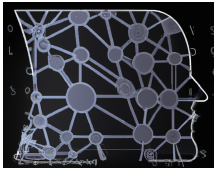
Example: embedded method (Lasso) regularizing model params with $L1$ penalty enables “automatic" feature selection:

$$\mathcal{R}_{\text{reg}}(\boldsymbol{\theta}) = \mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) + \lambda \|\boldsymbol{\theta}\|_1 = \sum_{i=1}^{n} \left(y^{(i)} - \boldsymbol{\theta}^\top \mathbf{x}^{(i)}\right)^2 + \lambda \sum_{j=1}^{p} |\theta_j|$$

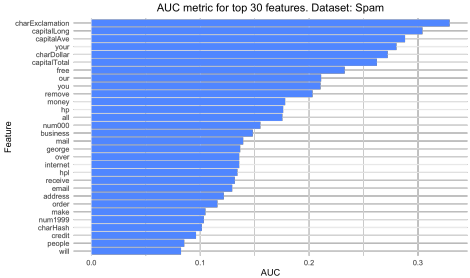


Filtering

INTRODUCTION

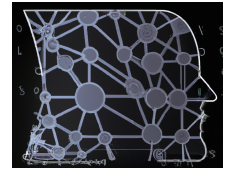


- **Filter methods** construct a measure that quantifies the dependency between features and the target variable.
- They yield a numerical score for each feature x_j, according to which we rank the features.
- They are model-agnostic and can be applied generically.



Exemplary filter score ranking for Spam data.

PEARSON & SPEARMAN CORRELATION



Pearson correlation $r(x_j, y)$:

- For numeric features and targets only.
- Measures linear dependency.
- $r(x_j, y) = \frac{\sum_{i=1}^{n}(x_j^{(i)} - \bar{x}_j)(y^{(i)} - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_j^{(i)} - \bar{x}_j)}\sqrt{(\sum_{i=1}^{n} y^{(i)} - \bar{y})}}, \qquad -1 \leq r \leq 1$

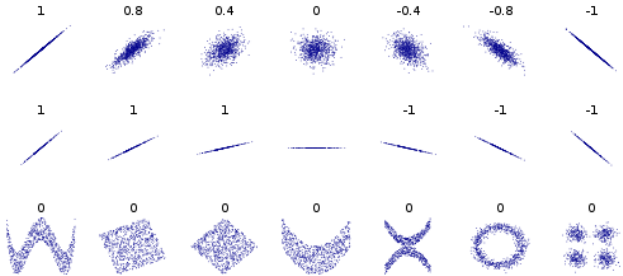
Spearman correlation $r_{SP}(x_j, y)$:

- For features and targets at least on ordinal scale
- Equivalent to Pearson correlation computed on ranks
- Assesses monotonicity of relationship

Use absolute values $|r(x_j, y)|$ for feature ranking:
higher score indicates a higher relevance.

PEARSON & SPEARMAN CORRELATION

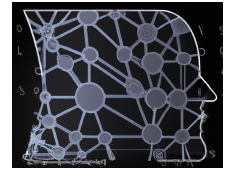
Only **linear** dependency structure, non-linear (non-monotonic) aspects are not captured:



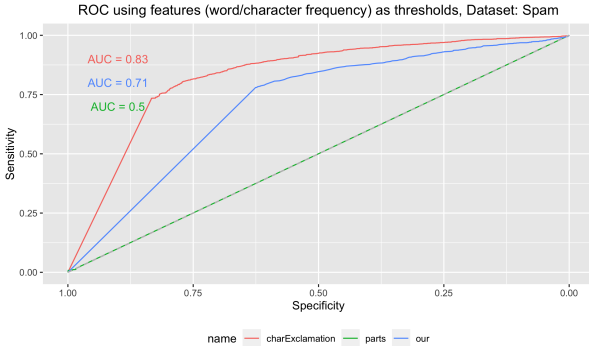
Comparison of Pearson correlation for different dependency structures.

To assess strength of non-linear/non-monotonic dependencies, generalizations such as **distance correlation** can be used.

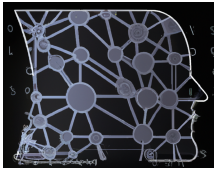
AUC/ROC



- For binary classification with $\mathcal{Y} = \{0, 1\}$ and numeric features
- Classify samples using single feature (with thresholds), compute AUC per feature as proxy for its ability to separate classes
- Features are then ranked; higher AUC scores $\rightarrow$ higher relevance.



FURTHER CRITERIA/METHODS



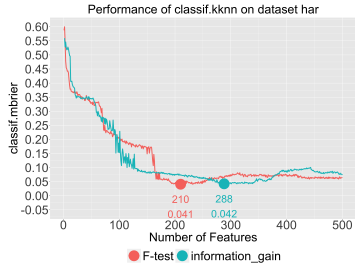
- Classification
  - χ^2-statistic
  - Welch's t-Test
  - F-Test
- Regression & Classification
  - Mutual Information
  - Relief
- Model-based
  - Tree-based feature importance (decision tree, extra trees, random forest)
  - Coefficients of linear model

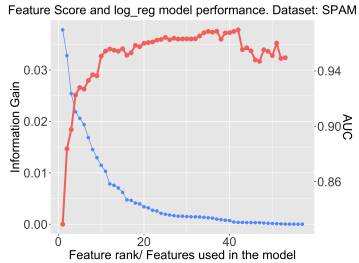
USING FILTER METHODS

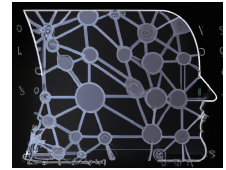
1. Calculate filter score for each feature x_j
2. Rank features according to score values.
3. Choose $\tilde{p}$ best features
4. Train model on $\tilde{p}$ best features.

How to choose $\tilde{p}$?

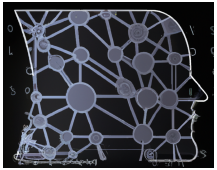
- Could be prescribed by the application
- Eyeball estimation: read from filter plots
- Treat as hyperparameter and tune in a pipeline, based on resampling







ADVANTAGES AND DISADVANTAGES



Advantages

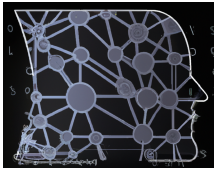
- Fast
- Typically scales well with the number of features p
- Generally interpretable
- Can be used with any inducer

Negative

- Does not take specifics of the inducing algorithm into account
- Redundant features will have similar weights
- Often univariate

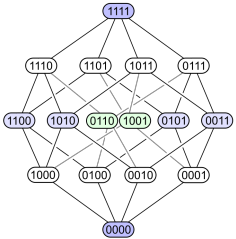
Wrapper

INTRODUCTION



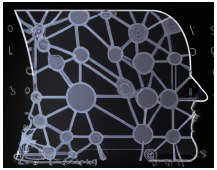
- Wrapper methods emerge from the idea that different sets of features can be optimal for different learners
- Wrapper is a discrete search strategy for S, where objective criterion is test error of learner as function of S. Criterion can also be calculated on train set, approximating test error (AIC, BIC)

$\Rightarrow$ Use the learner to assess the quality of the feature sets



Hasse diagram illustrating search space. Knots are connected if Hamming distance = 1
(Source: Wikipedia)

OBJECTIVE FUNCTION

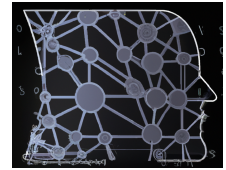


Given p features, **best-subset selection problem** is to find subset $S \subseteq \{1, \dots p\}$ optimizing objective $\Psi : \Omega \to \mathbb{R}$:

$$S^* \in \arg\min_{S \in \Omega} \{\Psi(S)\}$$

- Ω = search space of all feature subsets $S \subseteq \{1, \dots, p\}$. Usually we encode this by bit vectors, i.e., $\Omega = \{0, 1\}^p$ (1 = feat. selected)
- Objective Ψ can be different functions, e.g., AIC/BIC for LM or cross-validated performance of a learner
- Poses a discrete combinatorial optimization problem over search space of size = 2^p, i.e., grows exponentially in p (power set)
- Unfortunately can not be solved efficiently in general (NP hard; see, e.g., Natarajan, 1995)
- Can avoid searching entire space by employing efficient search strategies, traversing search space in a “smart" way

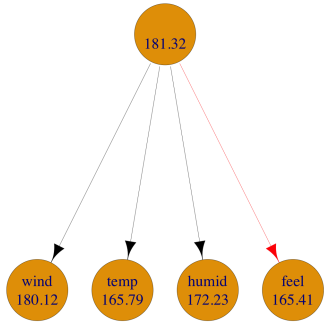
GREEDY FORWARD SEARCH



Let $S \subset \{1, \ldots, p\}$ be subset of feature indices.

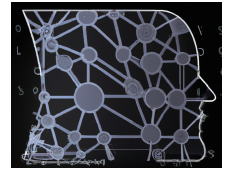
1. Start with the empty feature set $S = \emptyset$
2. For a given set S, generate all $S_j = S \cup \{j\}$ with $j \notin S$.
3. Evaluate the classifier on all S_j and use the best S_j

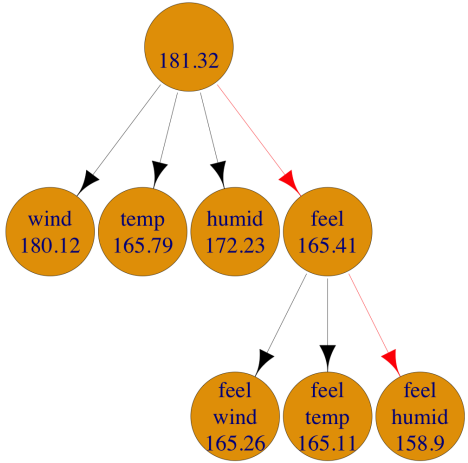
Example GFS on a subset of bike sharing data with features windspeed, temp., humidity and feeling temp. Node value is RMSE.



VISUALIZATION OF GFS

4. Iterate over this procedure





VISUALIZATION OF GFS

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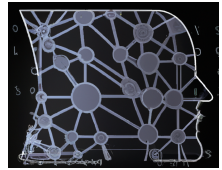
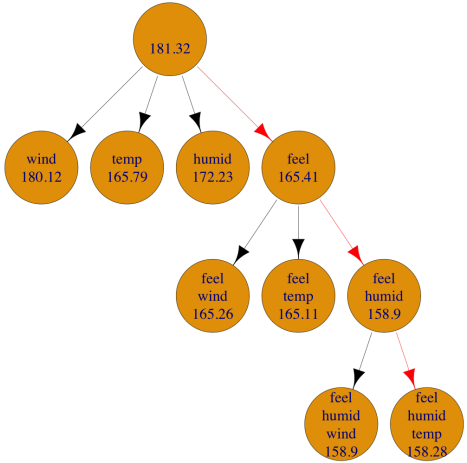
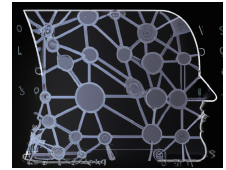
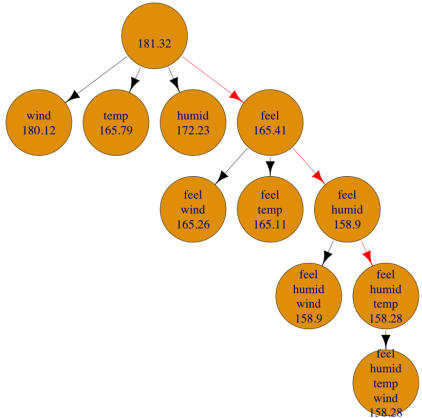




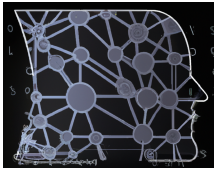
VISUALIZATION OF GFS





5. Terminate if performance does not improve further or max. number of features is used

GREEDY BACKWARD SEARCH

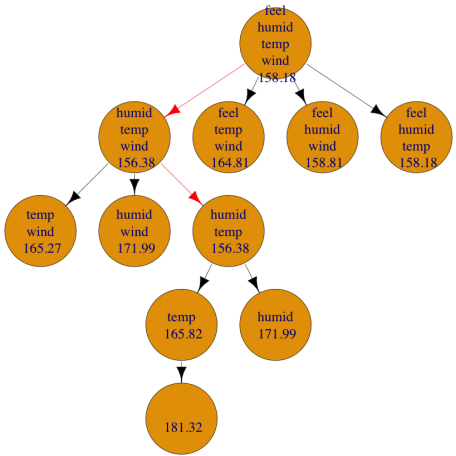


- Start with the full index set of features $S = \{1, \ldots, p\}$.
- For a given set S generate all $S_j = S \setminus \{j\}$ with $j \in S$.
- Evaluate the classifier on all S_j and use the best S_j.
- Iterate over this procedure.
- Terminate if:
  - the performance drops drastically, or
  - falls below given threshold.

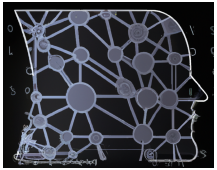
- GFS is much faster and generates sparser feature selections
- GBS much more costly and slower, but sometimes slightly better.

VISUALIZATION OF GBS

Example Greedy Backward Search on bike sharing data



ADVANTAGES AND DISADVANTAGES



Advantages

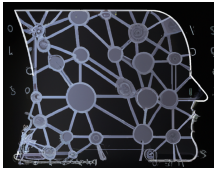
- Inducer-agnostic
- Any performance measure can be used
- Optimizes the desired performance measure directly

Disadvantages

- Expensive
- Does not scale well with the number of features
- Does (in general) not use additional info about model structure
- Nested resampling becomes necessary

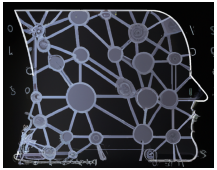
Further Selection Techniques

EMBEDDED FEATURE SELECTION



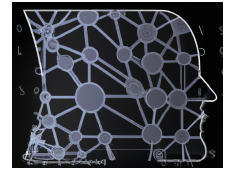
- Select features during the learning process
- Explicitly – via regularization
  - LASSO/L1
  - Requires a solver that can set weights to zero
- Implicitly – by construction of the algorithm
  - Decision-tree based algorithms
  - Can ignore irrelevant features by not selecting them (no guarantee, though)

DOMAIN KNOWLEDGE

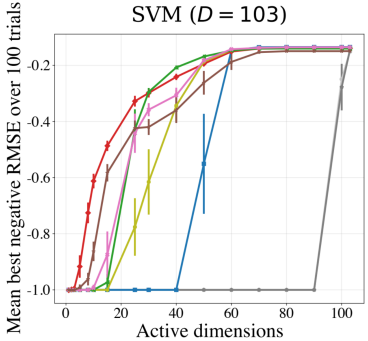


- ID columns
  - Often ordered, can leak time information
  - If they identify a person etc, replace by features describing that person
  - Contains no information that can generalize
- Duplicate features
  - Cause issues for (linear) models
  - Slow down learning
  - Reduce model interpretability

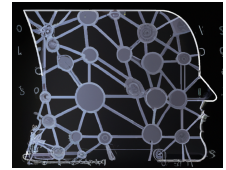
MULTI-OBJECTIVE OPTIMIZATION OF FEATURES



- Optimize two competing Objectives:
  1. goodness-of-fit
  2. number of variable
- Use global optimization algorithm
  - Evolutionary Algorithm
  - Sparse Bayesian optimization



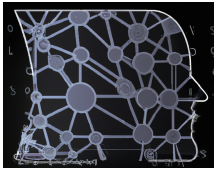
BORUTA (1)



- All-relevant feature selection
  Goal: find all features that have affect the prediction
  $\rightarrow$ Can include redundant features
- Idea: use *shadow variables* to contrast existing features against
- Method: Extend a feature scoring to an iterative testing mechanism

Boruta for those in a Hurry by Miron B. Krusa

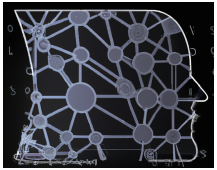
BORUTA (2)



1. Create a copy of each feature, shuffle these copies
2. Fit a random forest on the new dataset
3. An attribute is deemed important, if its feature importance is higher than the maximal importance of all randomised attributes
4. Repeat steps 1-3 for N iterations
5. Execute SHT with the null hypothesis that the importance of a feature is equal to the maximal importance of all randomised attributes
6. Repeat steps 1-5 until the feature set is stable

Wrap-up

TAKEAWAYS



- There can be multiple goals in feature selection:
  - Single Feature Selection
  - Multiple Feature Selection
  - All-relevant Feature Selection
- Important step for good prediction quality and fast models
- Different feature selection methods have different goals and use different mechanisms $\rightarrow$ model selection problem